Neural correlates of the word-onset response in human iEEG

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Introduction

The word-onset response in free recall

Most intracranial EEG (iEEG) studies of free recall contrast words that are later recalled with words that are forgotten (the subsequent memory effect, SME), or study the retrieval period. As both conditions contain a word, this contrast measures how responses differ with later memory, but cancels out the response to the word itself. Only a few studies have examined the word-onset response directly. Kucewicz et al. (2019) showed that high gamma (HG) activity after word onset moves from visual cortex to the medial temporal lobe (MTL) and then to prefrontal cortex, and Marks et al. (2021) found that theta, alpha/beta and gamma activity each spread across regions on their own timelines during word encoding.

High gamma and alpha as markers of cortical engagement

Broadband HG (70–150 Hz) is widely used as a measure of local population spiking (Manning et al. 2009; Ray & Maunsell 2011) and is better described as a broadband signal than as an oscillation (Burke, Ramayya & Kahana 2015). At stimulus onset, alpha and beta power decrease (event-related desynchronization; Pfurtscheller & Lopes da Silva 1999), which is thought to release cortex from alpha-mediated inhibition (Jensen & Mazaheri 2010). An alpha/beta decrease together with an HG increase is the typical cortical activation response, first described in sensorimotor cortex (Crone et al. 1998a, b) and observed in visual cortex (Lachaux et al. 2005), including for pseudowords and consonant strings (Vidal et al. 2010). In successful memory encoding, words later recalled show lower low-frequency power and higher HFA than words later forgotten (Sederberg et al. 2003; Long et al. 2014). This broadband tilt also appears outside memory, during arithmetic (Randazzo et al. 2019, preprint), where it predicts response speed, mainly in frontal regions (DeHaan et al. 2025, preprint). Not every region is engaged by a stimulus, however – alpha increases in task-irrelevant cortex are thought to actively inhibit it (Jensen & Mazaheri 2010), and HG often decreases during tasks in default-mode regions (Ossandón et al. 2011).

Timing of visual word processing

Reading a word unfolds in stages: visual features are extracted within about 100 ms (Thorpe et al. 1996), the letter string is recognised as a word at about 200 ms (Cohen et al. 2000), its sound is activated automatically (Van Orden 1987), and its meaning is integrated with context at about 400 ms (Kutas & Hillyard 1980). These stages map onto a reading network that runs from occipital cortex through left fusiform gyrus, which contains the visual word form area (VWFA; Cohen et al. 2000; Dehaene & Cohen 2011), to temporal, supramarginal and inferior frontal cortex (Price 2012; Taylor et al. 2013), with posterior responses to letter strings preceding more anterior responses to lexical status (Nobre et al. 1994; Allison et al. 1994).

iEEG resolves when each part of this network responds. Gamma activity begins at about 150 ms in basal temporal cortex, reaches the VWFA at about 200 ms, and reaches superior temporal and Broca's areas after 400 ms (Mainy et al. 2008); word-selective ventral temporal electrodes respond at about 150 ms (Kadipasaoglu et al. 2016); and the effects of word properties move anteriorly along the ventral stream, with feedback to posterior regions at 250–500 ms (Woolnough et al. 2021). Beyond the ventral stream, MEG shows responses proceeding from temporal cortex to Broca's area and finally to orbitofrontal cortex at about 370 ms (Halgren et al. 2002), and MTL neurons respond later still, at about 270–400 ms (Mormann et al. 2008). Early visual responses are sharp, whereas later responses are broad, consistent with feedforward input followed by feedback (Halgren et al. 2006; Ghuman et al. 2014). However, most intracranial studies of reading record from tens of patients with sparse, clinically determined coverage, and focus on the ventral stream, so the order in which regions across the whole brain respond to a word has not been measured within one large dataset.

Synchrony between regions

Beyond local power, word onset may change how regions synchronize. Alpha phase synchrony is thought to coordinate distant regions (Palva & Palva 2007), and alpha power can fall while alpha synchrony between regions rises (Hanslmayr et al. 2016). In free recall, slow-frequency synchrony increases while high-frequency activity desynchronizes during successful encoding and retrieval (Burke et al. 2013; Solomon et al. 2017), and theta hubs mark successful retrieval in this dataset (Rao et al. 2025). Connectivity measured at stimulus onset is harder to interpret than power: an evoked response shared across regions can raise phase and envelope coupling without any interaction between them, and subtracting the average evoked response does not fully remove it, because its amplitude and latency vary across trials (Truccolo et al. 2002; David et al. 2006). We therefore treat synchrony as a secondary analysis.

Temporal expectation

Reaction times and neural responses depend on how long the participant waits for a stimulus and on the changing probability that it will appear (the hazard rate; Niemi & Näätänen 1981; Nobre & van Ede 2018). Temporally expected events evoke larger visual responses: in monkey V1, gamma increases and alpha decreases when a stimulus is expected (Lima et al. 2011), and expectation speeds and enhances visual processing in humans (Doherty et al. 2005).

Predicting recall from the word-onset response

Spectral features recorded while a word is studied predict whether it will be recalled (L2-regularised logistic regression; Ezzyat et al. 2017, 2018), although these classifiers partly track general fluctuations in brain state rather than encoding of the item (Weidemann & Kahana 2021). Activity before a stimulus can also predict memory: pre-stimulus hippocampal theta predicts later recognition but not recall (Merkow et al. 2014), and pre-stimulus MTL and cortical low-frequency activity predict encoding success in iEEG (Fell et al. 2011). Comparing recalled with forgotten words region by region, HFA predicting later recall appears first in visual, fusiform and medial temporal regions (about 0–500 ms) and later in left inferior frontal, posterior parietal and ventrolateral temporal cortex (about 400–900 ms; Burke, Long et al. 2014). These approaches either pool features across the brain and the whole word period, or test each region and time point for a difference in mean activity; how well each region's single-trial response predicts recall, and when that prediction arises, has not been measured with classifiers.

The present study

There have been few whole-brain maps of the response to word onset itself, contrasting each word to a pre-stimulus baseline. With 362 patients, we obtain a complete set of HG onset latencies across all regions, extending the existing literature. We also test phase (alpha ciPLV) and envelope (HG AEC-c) coupling in the same data.

Methods

Data and cohort

- OpenNeuro BIDS releases of the RAM / Penn free-recall data (FR1, catFR1) and pyFR, bipolar montage. OpenNeuro ds004789 (FR1), ds004809 (catFR1), ds004865 (pyFR).
- 362 subjects, 1009 sessions: FR1 232 subj / 466 sess, catFR1 212 / 465, pyFR 38 / 78.
- Bipolar channels whose contacts are > 20 mm apart, and flat channels are dropped.
- Exclusions: missing / < 499 Hz sampling rate, denylist / manual, mains-harmonic peak $> 5\times$ its neighbours at a non-notched frequency (9 sessions, all 100 Hz peaks at 60 Hz sites).
- Contrast: **word on** (0–600 ms after word onset) vs **word off** (-700 to -100 ms, the blank screen before the same word). Words on screen 1600 s; blank between words ~750-1250 ms.

Preprocessing

- Word-off and word-on clips loaded separately per word, each with 500 ms of real data buffer on both sides.
- Resampled to 500 Hz, notch 60 Hz + 120 Hz harmonic (4th-order Butterworth, ± 2 Hz); buffer cropped before every estimate. (Widmann et al. 2015; de Cheveigné & Nelken 2019; Gramfort et al. 2013; Virtanen et al. 2020)
- Why 500 ms buffer: resampling (FFT) and notching each short clip ring at its edges for > 50 ms; the ringing is simultaneous across channels and biased the amplitude-correlation measure (AEC-c). No effect on multitaper alpha (Supplement)

Electrode localization

- Bipolar pairs labelled by lab localization default (depth: volumetric; grid/strip: surface). (Desikan et al. 2006; Fischl et al. 2002, 2004; Avants et al. 2008; Wang et al. 2013; Yushkevich et al. 2015; Azarion et al. 2014 (check); bipolar: Mercier et al. 2022; Li et al. 2018)
- 12 Burke ROIs: frontal, temporal, parietal, occipital, limbic, hippocampus $\times$ hemisphere; sub regions = anatomical labels within them. (Burke et al. 2013)

Power

- Alpha (8–13 Hz) and 5–100 Hz spectrum: multitaper, 2 Hz bandwidth, 600 ms windows; spectrum in 10 log-spaced bins (48–52 / 58–62 Hz excluded). (Thomson 1982; Slepian 1978; Percival & Walden 1993; Prerau et al. 2017)
- High gamma (70–150 Hz): Hilbert in 8×10 Hz sub-bands, each divided by its word-off mean, averaged (broadband, every frequency weighted equally). (Edwards et al. 2009 (check); Bruns 2004; Manning et al. 2009; Ray & Maunsell 2011)
- Per electrode: Cohen's d of per-trial power, word on vs word off. (Cohen 1988; Lakens 2013)

High-gamma latency

- Responsive electrodes: per-trial high-gamma power word on > word off (one-tailed Welch t test, BH-FDR within session): 5004 of 122,339 electrode-sessions, 293 subjects. (Welch 1947; Benjamini & Hochberg 1995; selection-then-measure: Kriegeskorte et al. 2009)
- Latency: trial-mean envelope minus word-off mean, 30 ms smoothing, first time it reaches half its peak within 0–800 ms (half-maximum). (Kiesel et al. 2008; Liesefeld 2018; filter smearing: VanRullen 2011)
- Subject median per region → mean ± SEM across subjects; 12 ROIs and subregions (hemispheres pooled).

Synchrony

- Alpha: ciPLV (MNE `spectral_connectivity_epochs`, multitaper, 2 Hz bandwidth). (Bruña et al. 2018; Nolte et al. 2004; MNE-Connectivity (Zenodo))
- High gamma: AEC-c (70–110 Hz band-pass + Hilbert, `envelope_correlation`, pairwise orthogonalized, signed, mean over trials); 70–110 Hz so the envelope stays below the band (Bedrosian). (Hipp et al. 2012; Bedrosian 1963)
- Pairs sharing a contact excluded. (Mercier et al. 2017) Per electrode: mean on − off change over all partners (Rao et al. 2025).

Time-resolved network (epochs)

- 9 overlapping 200 ms epochs every 100 ms, 0–1000 ms (stops before the word's offset response at 1600 ms), each minus the mean of three 200 ms word-off epochs.
- Region pairs with ≥ 100 subjects. Hub (Rao): a region's mean change over its connections; t across subjects; two-stage FDR (Benjamini–Krieger–Yekutieli) over region × epoch. (Benjamini et al. 2006; Rubinov & Sporns 2010)
- Edges: each significant hub's 5 strongest connections (|mean change|), either sign; width ∝ |t| (Solomon et al. 2017 style).

Mixed-effects models (R lme4 / emmeans; `mixed_models.R`)

- Latency: latency_ms ~ lobe + (1|sub) + (1|sub:lobe) + (1|sub:region), one row per responsive electrode; all 15 lobe pairs, Holm.
- Power: cohens_d ~ lobe × band + (1 + band|sub) + (1|sub:lobe) + (1|sub:lobe:band) + (1|sub:region) + (1|sub:label); lobe × band interaction contrast per lobe pair, Holm.

Anticipation

- Responsive electrodes (as latency); trials split at the session's median blank length; peak (0–800 ms) of the trial-mean high-gamma response after long minus short waits; subject mean per ROI, one-sample t, BH-FDR over ROIs.

Recall prediction (encoding contrast: recalled vs not recalled)

- High-gamma response (as latency) in 100 ms windows, 0–1000 ms, z-scored within session; per subject × lobe (≥ 5 electrodes) × window: L2 logistic regression (C = 7.2e-4, Ezzyat et al. 2017), balanced class weights, leave-one-session-out (single-session subjects: leave-one-list-out); AUC of pooled held-out predictions.

- Across subjects: one-sided t of AUC > 0.5, BH-FDR over lobe $\times$ window; permutation chance ($\sim$0.49) reported only.

Reliability

- Subjects' sessions split at random into two halves (≥ 2 sessions per half; riley-thesis rules); per half, d (power) or node strength (synchrony) averaged per anatomical region; test-retest = Pearson r across the subject's regions between halves.

Statistics

- Sessions averaged per electrode, electrodes per subject $\times$ region (≥ 1 electrode); one-sample t vs 0 across subjects; BH-FDR within each figure / band; ≥ 30 subjects per region. (Holmes & Friston 1998; Benjamini & Hochberg 1995; Seabold & Perktold 2010)

Validation (simulations, real montages and events, signal replaced)

- Power: planted high-gamma gain recovered ($\approx$ identity without noise), 0 elsewhere.
- ciPLV: lagged alpha coupling recovered monotonically; 0 for null, zero-lag leak, envelope plant; planted 400–800 ms coupling found only at 400–800 ms.
- AEC-c: planted shared envelope recovered ($0.16 \rightarrow 0.39 \rightarrow 0.59$ with depth); non-target pairs 0; zero-lag leak rejected; timing correct.
- Spectrum: 60 Hz notch ringing at clip edges (resolved with the 500 ms buffer; 64 Hz bin now unbiased).

Results

1. Power changes at word onset

- Alpha: decrease in occipital (L -0.14, R -0.11 d), smaller decreases in limbic, hippocampus, R temporal; increase in frontal (L $+0.031$, R $+0.024$) and L parietal ($+0.020$).
- High gamma: large increase in occipital (L $+0.41$, R $+0.33$ d); smaller increases in limbic ($+0.04$), hippocampus ($+0.04$), L temporal ($+0.02$); slight decrease R frontal (-0.011).

Spectral profile

- Frontal: theta/low-alpha increase (t up to 11) with beta decrease (t to -13.5).
- Occipital: alpha–beta decrease (10.6 Hz t ≈ -8), broadband increase ≥ 35 Hz (t up to 8.6).
- Limbic / hippocampus: mild alpha decrease, gamma increase.
- Occipital subregions (hemispheres pooled; the 4 with most subjects): fusiform (199), lateral occipital (86), lingual (77), cuneus (32). All: alpha/beta decrease and gamma increase ($\geq$35–47 Hz); fusiform strongest (t ≈ -7.5 at 11 Hz, $+8.9$ at 64 Hz); lateral occipital also a 6 Hz increase; cuneus weakest (gamma only). TFCE, family-wise $p < .05$. (Smith & Nichols 2009; Mensen & Khatami 2013; Nichols & Holmes 2002) #### Mixed model (lobe $\times$ band)

- Does the alpha-vs-high-gamma pattern differ between lobes (lobe $\times$ band interaction)? 11 of 15 pairs differ (Holm).
- Occipital differs from every other lobe (|d| 0.43–0.53); frontal from temporal, limbic and hippocampus (0.06–0.10); parietal from limbic and hippocampus; hippocampus from temporal.
- Not different: frontal vs parietal, parietal vs temporal, limbic vs temporal ($p = .07$), hippocampus vs limbic.

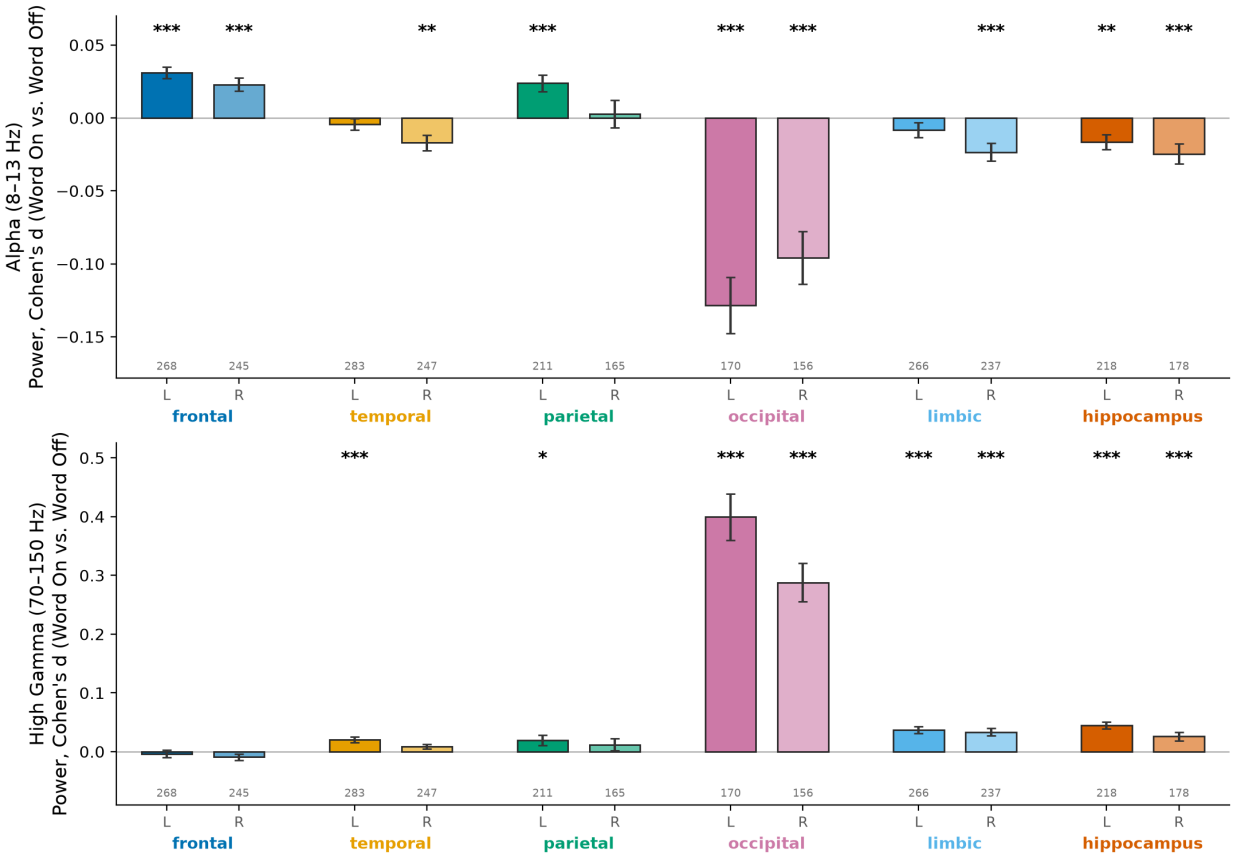


Figure 1: Power per ROI

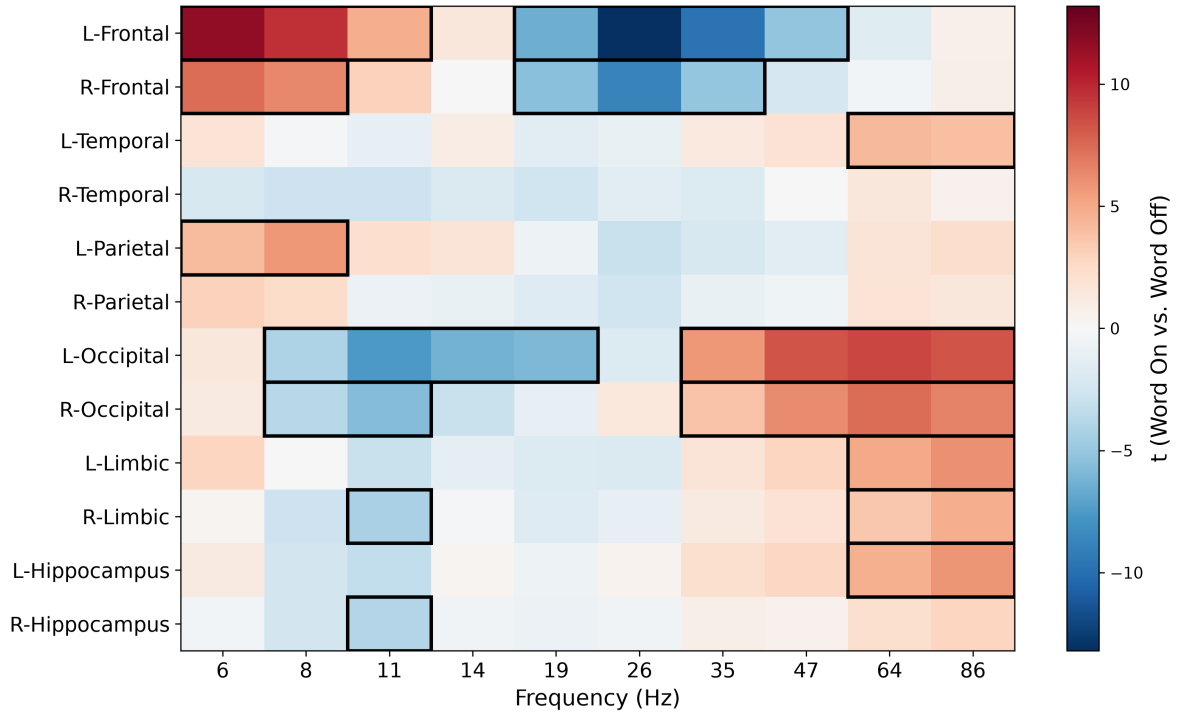


Figure 2: Spectrum heatmap

2. Order of activation

- Posterior → anterior: R occipital 128 ms, L occipital 157 → limbic, R temporal (~164–169) → R hippocampus, R parietal (~175–178) → L temporal 199, L hippocampus 207, R frontal 216 → L parietal 235, L frontal 259 ms.
- Subregions: lateral occipital 122 → lingual 150, fusiform 157 → parahippocampal 167, amygdala 170, insula 174 → superior parietal 188, temporal gyri ~189–194, hippocampus 192 → supramarginal 214, frontal gyri 215–237 → orbitofrontal 300 ms.
- Early regions: sharp peak ~150 ms; hippocampus / insula / frontal: broad peaks ~350–550 ms.

Mixed model (latency per lobe)

- Estimated latency: occipital 148 → limbic 171 → temporal 190 → hippocampus 199 → parietal 215 → frontal 241 ms.
- Neighbouring steps (brackets): occipital < limbic ($p = .045$) and parietal < frontal ($p = .045$); limbic–temporal ($p = .08$), temporal–hippocampus and hippocampus–parietal ($p = .39$) not separable.
- All 15 pairs (CSV): 12 differ (Holm $p < .05$); posterior → anterior holds at the lobe level, not every step.

3. Synchrony

- Alpha ciPLV increases in 10 / 12 ROIs, about twice as much in occipital (~0.0016) as elsewhere (~0.0003–0.0008).

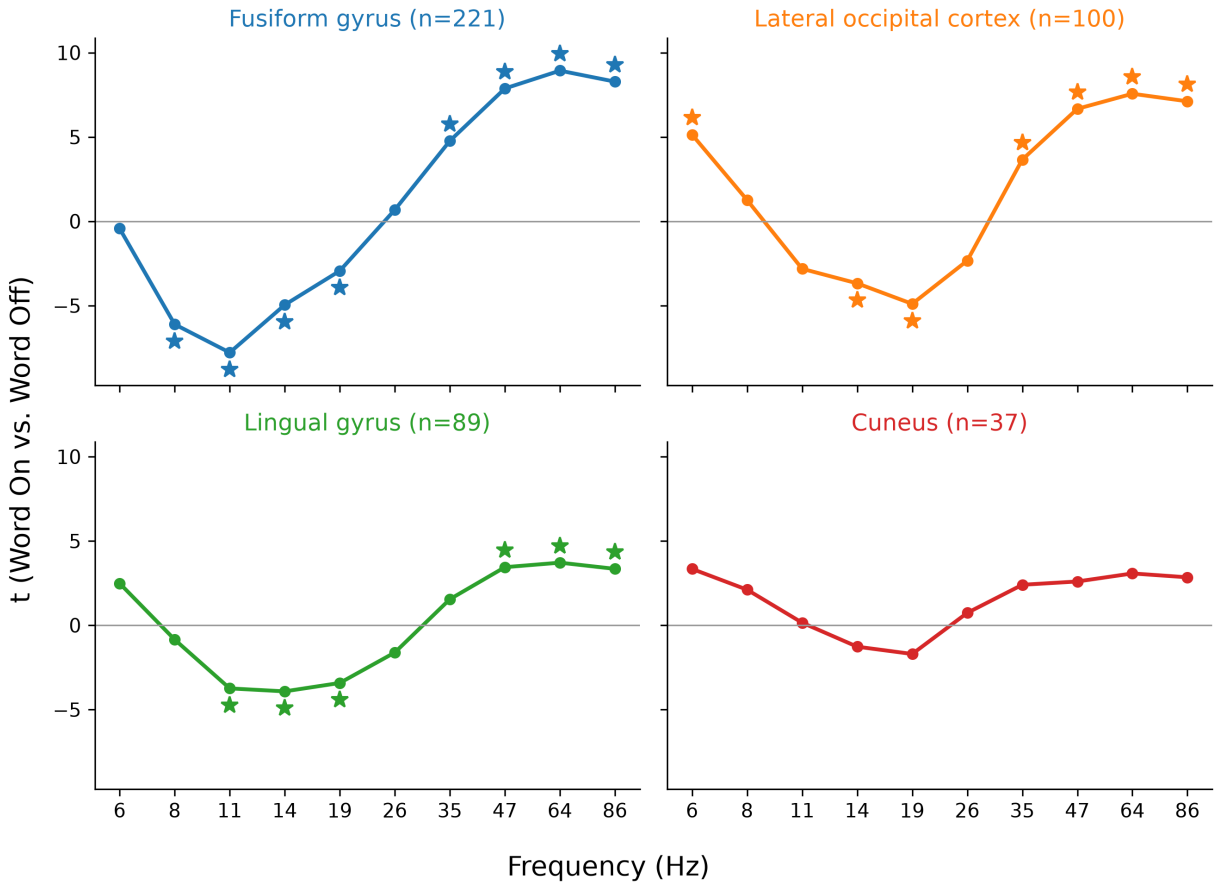


Figure 3: Occipital case studies

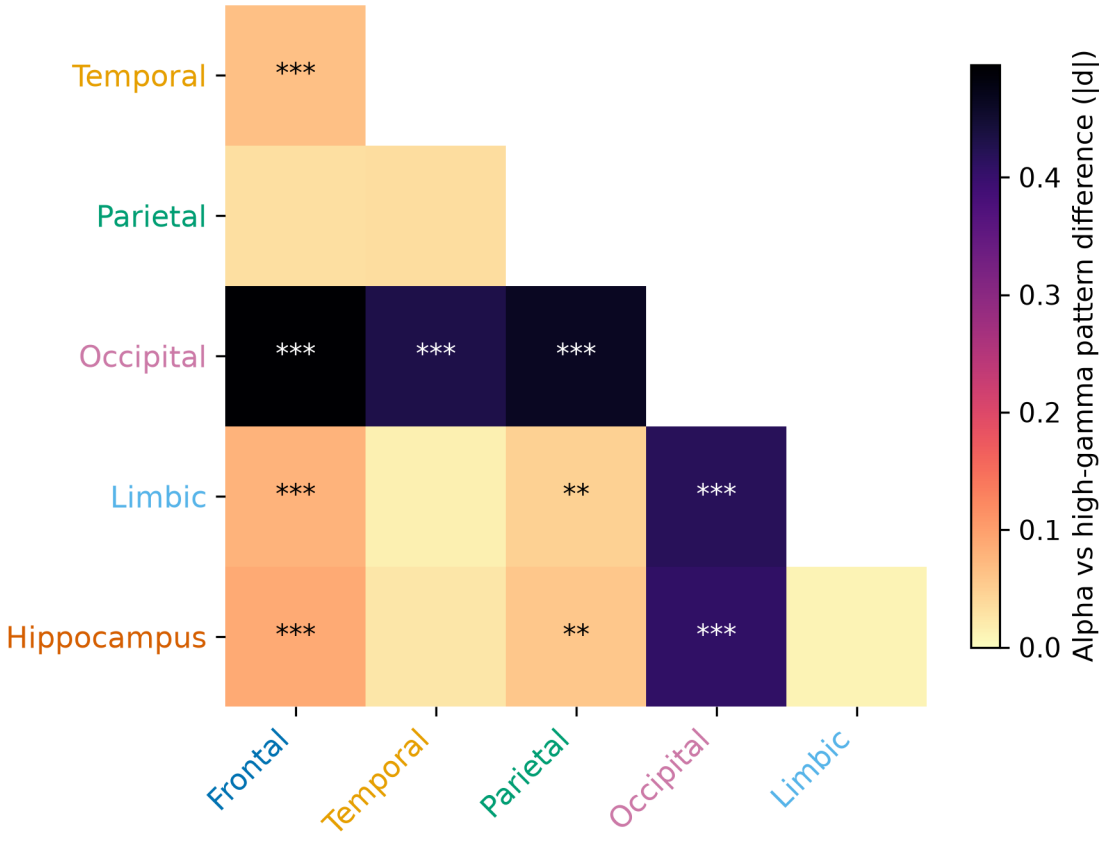


Figure 4: Power, mixed model

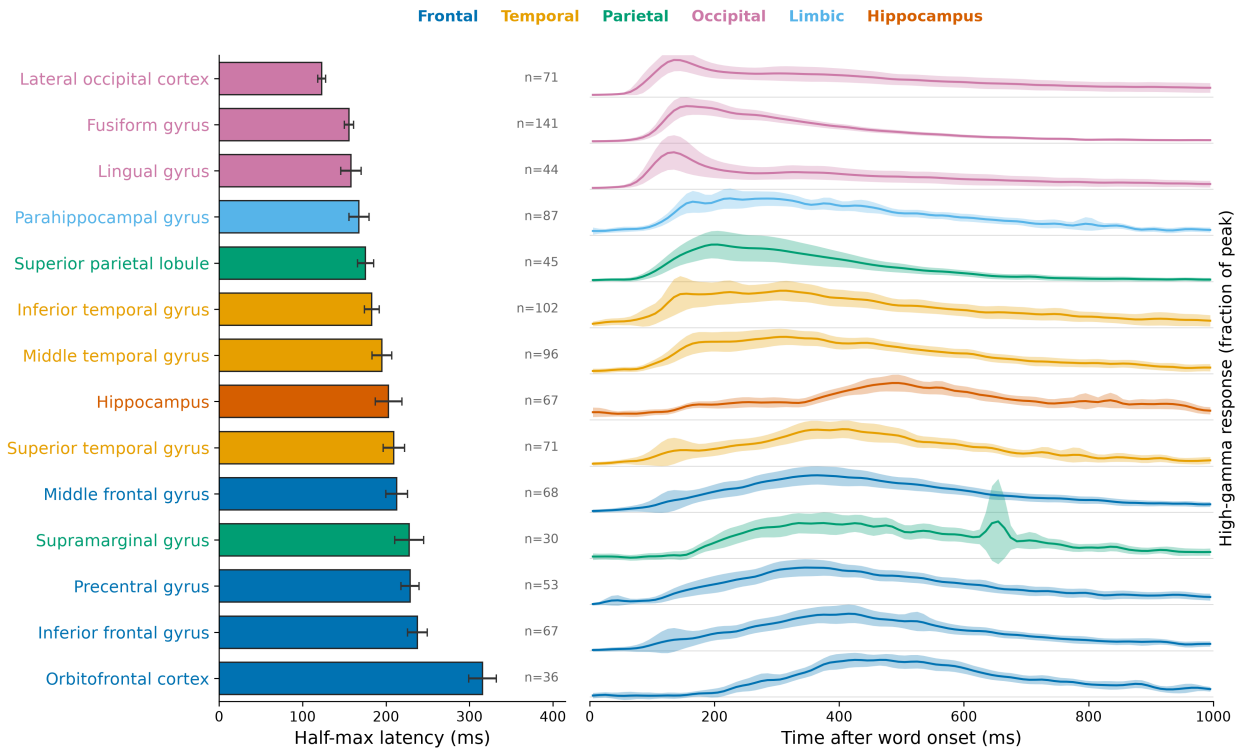


Figure 5: Latency, subregions

- High-gamma AEC-c: no significant change in any ROI.

Time-resolved network

- Alpha: 38 hub × epoch cells; occipital hubs 0–400 ms; parietal from 100 ms; frontal, temporal, limbic mainly 400–1000 ms; fine regions (9 cells): L fusiform 0–400 ms → L inferior/middle temporal 500–1000 ms, L orbitofrontal 400–600 ms.
- High gamma: 3 hubs (bilateral occipital 0–200 ms, L limbic 200–400 ms); no fine-region hubs; no late desynchronization.

4. Anticipation (blank length before the word)

- Occipital high-gamma peak larger after long than short waits (L +0.078, R +0.053; q = .024 both); no other ROI significant.
- Consistent with temporal expectation boosting visual responses (Lima et al. 2011, monkey V1; Nobre & van Ede 2018), but a shorter blank also leaves less time for the previous word's offset response to decay (baseline).

5. Recall prediction over the response (encoding contrast)

- No lobe predicts recall before 200–300 ms (first significant window: frontal and occipital 200–300 ms).
- Frontal: sustained 200–1000 ms, highest at 800–900 ms (AUC 0.525).

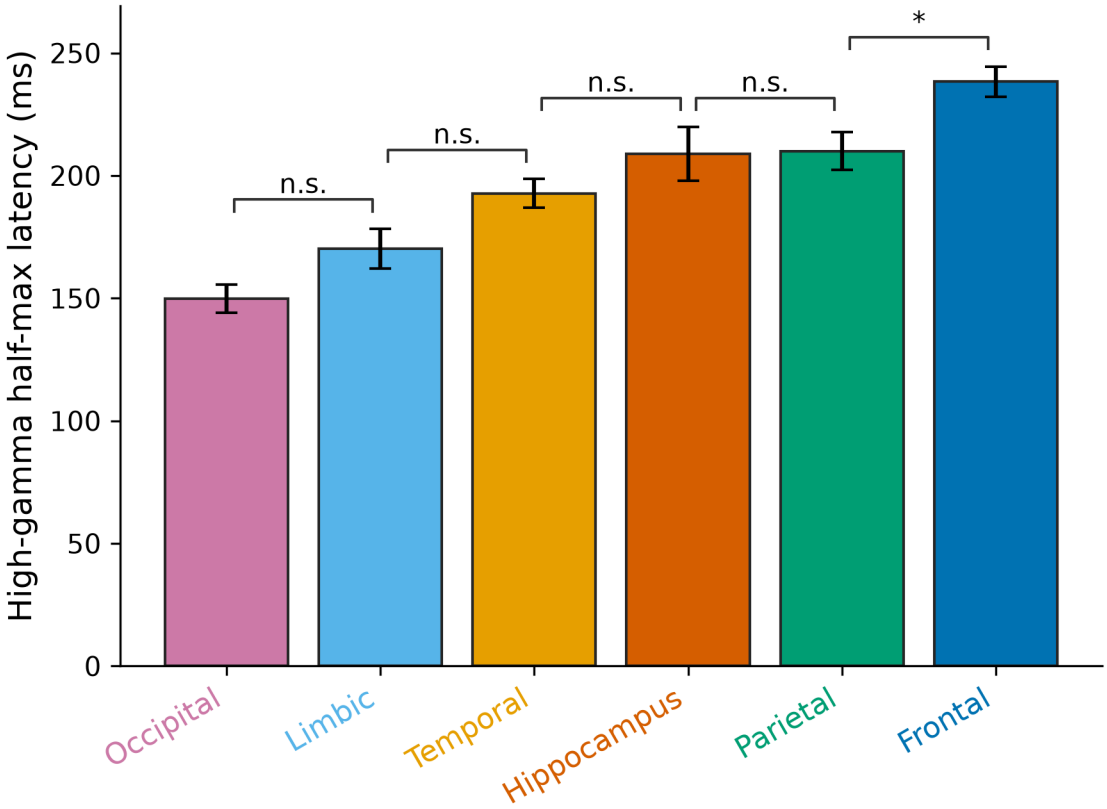


Figure 6: Latency, mixed model

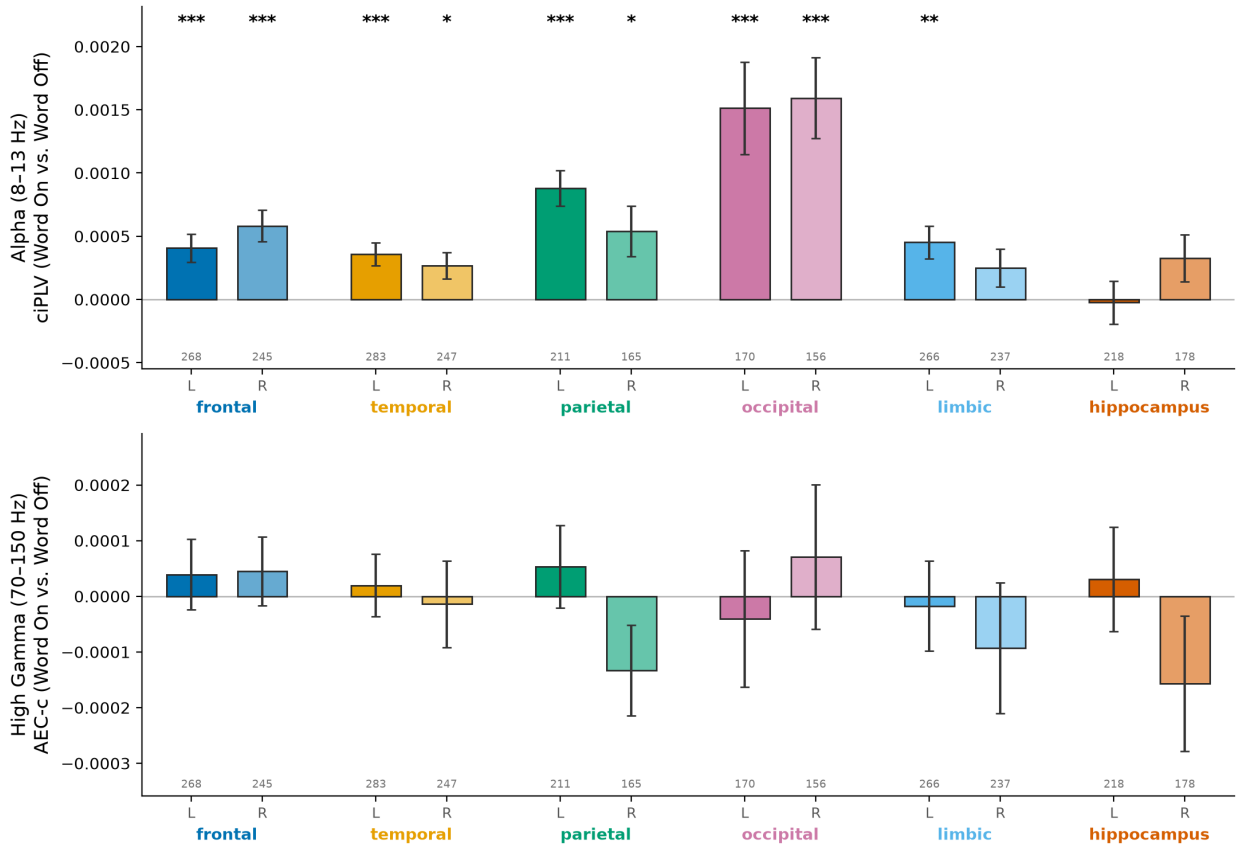


Figure 7: Synchrony per ROI

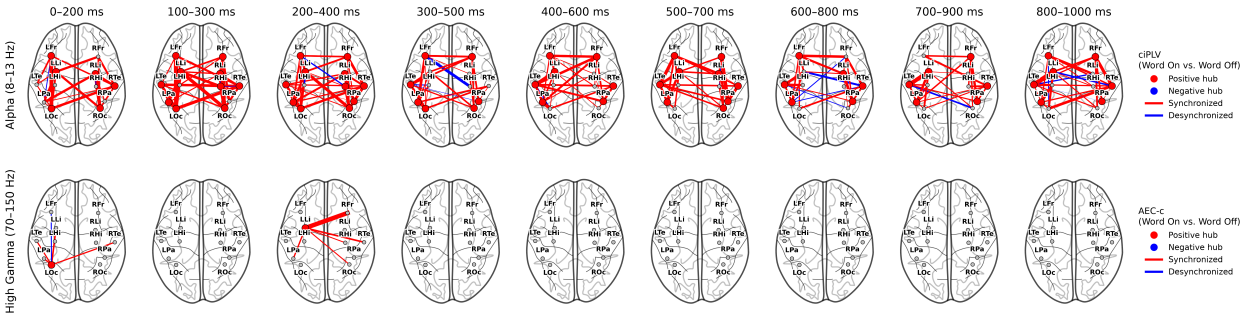


Figure 8: Epoch network

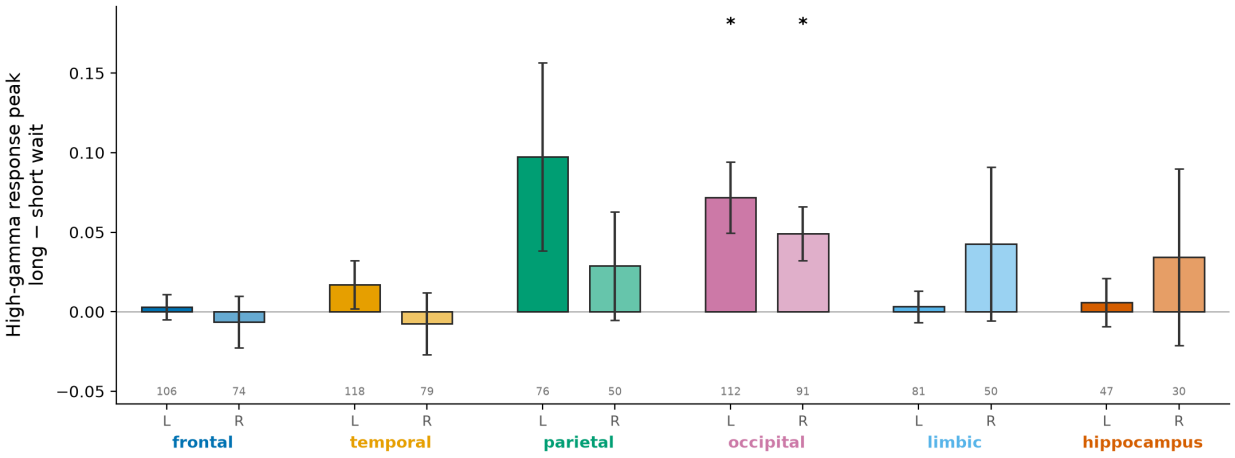


Figure 9: Anticipation

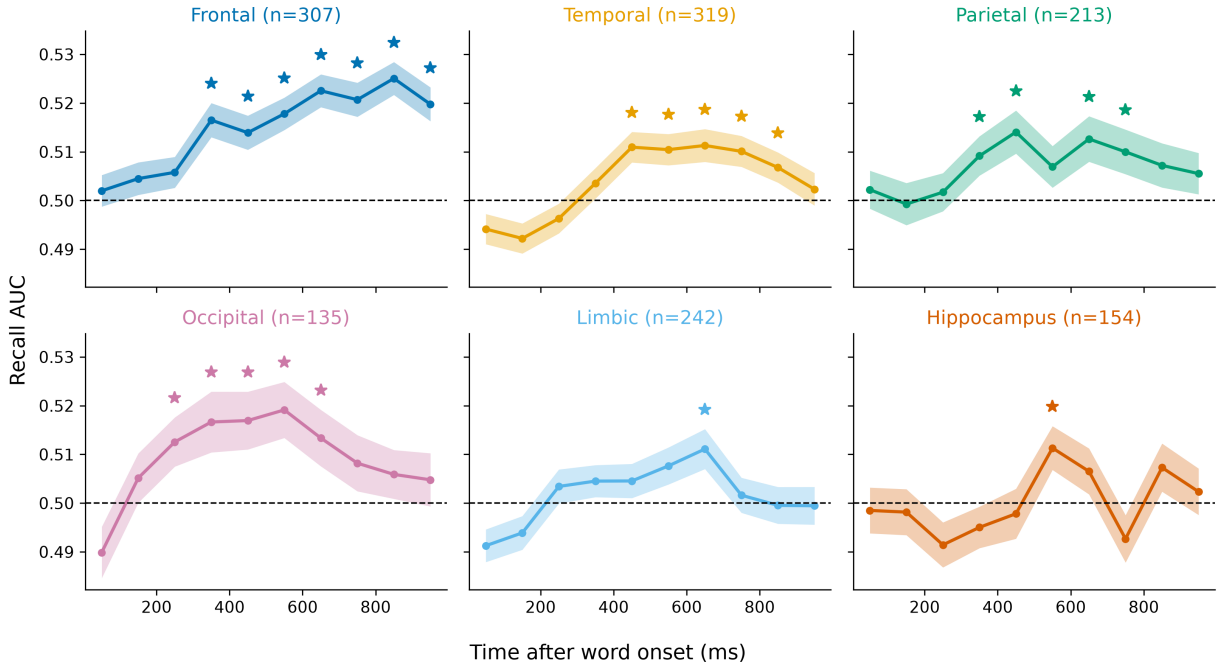


Figure 10: Recall decoding over time

- Occipital: 200–700 ms only (peak 0.521 at 300–400 ms); temporal 400–900 ms; parietal 300–900 ms (patchy); limbic and hippocampus 500–700 ms.
- AUCs are low (≤ 0.525) because each classifier uses one lobe's high gamma in one window. Same direction as Burke, Long et al. 2014 (encoding contrast: early posterior, late frontal HFA SME).

6. Reliability across sessions

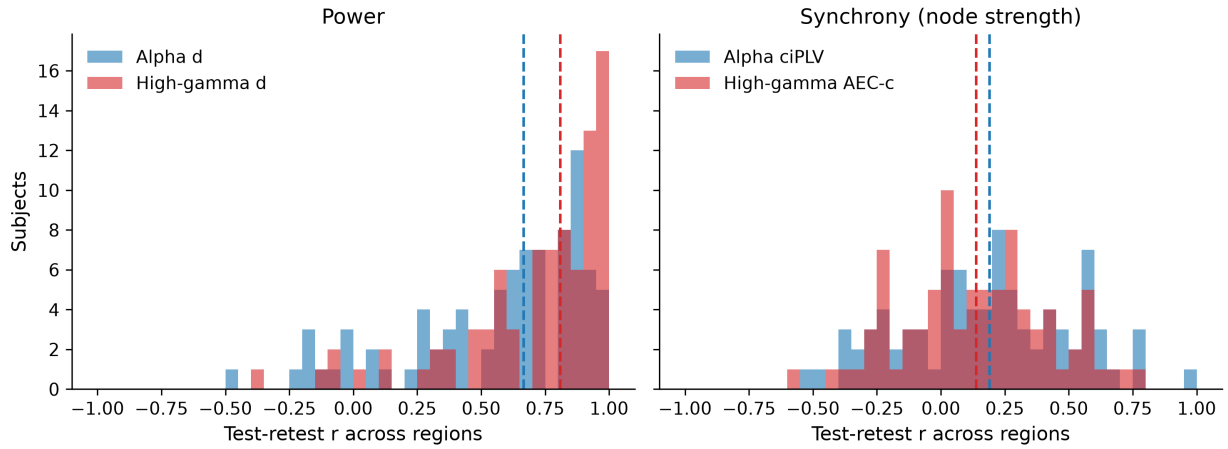


Figure 11: Reliability

- 87 subjects with ≥ 2 sessions per half. Power maps repeat: median r 0.63 (alpha), 0.79 (high gamma). Synchrony maps much less: 0.16 (alpha ciPLV), 0.13 (high-gamma AEC-c). **(verify: CSV from 2026-10-14, script changed since)**

Supplementary

Limitations / discussion

- Word-off baseline starts ~80-410 ms after the previous word's offset: possible offset response / anticipation in the baseline; the blank-length split (Results 4) cannot separate the two.
- No ERP subtraction: evoked activity contributes to power and synchrony, especially occipital.
- Neighbour-transferred labels (~11% / 21% disagreement in LOSO).
- High-gamma phase metrics avoided (broadband HFA has no meaningful phase); AEC-c null is consistent with high gamma being local and asynchronous (Solomon et al. 2017, 2019).
- Multiple comparisons corrected within figures, not across.

Stimulus-locked activity in connectivity

- Subtracting the average ERP leaves a residual because amplitude and latency vary across trials: Truccolo et al. 2002; Wang, Chen & Ding 2008; Knuth et al. 2006.
- Evoked and induced activity can't be cleanly separated: David et al. 2006; Shah et al. 2004. This supports the PI's position.
- Is alpha phase reset? For: Makeig et al. 2002; Hanslmayr et al. 2007. Against: Mazaheri & Jensen 2006. Hard to tell apart: Yeung et al. 2004; Sauseng et al. 2007.

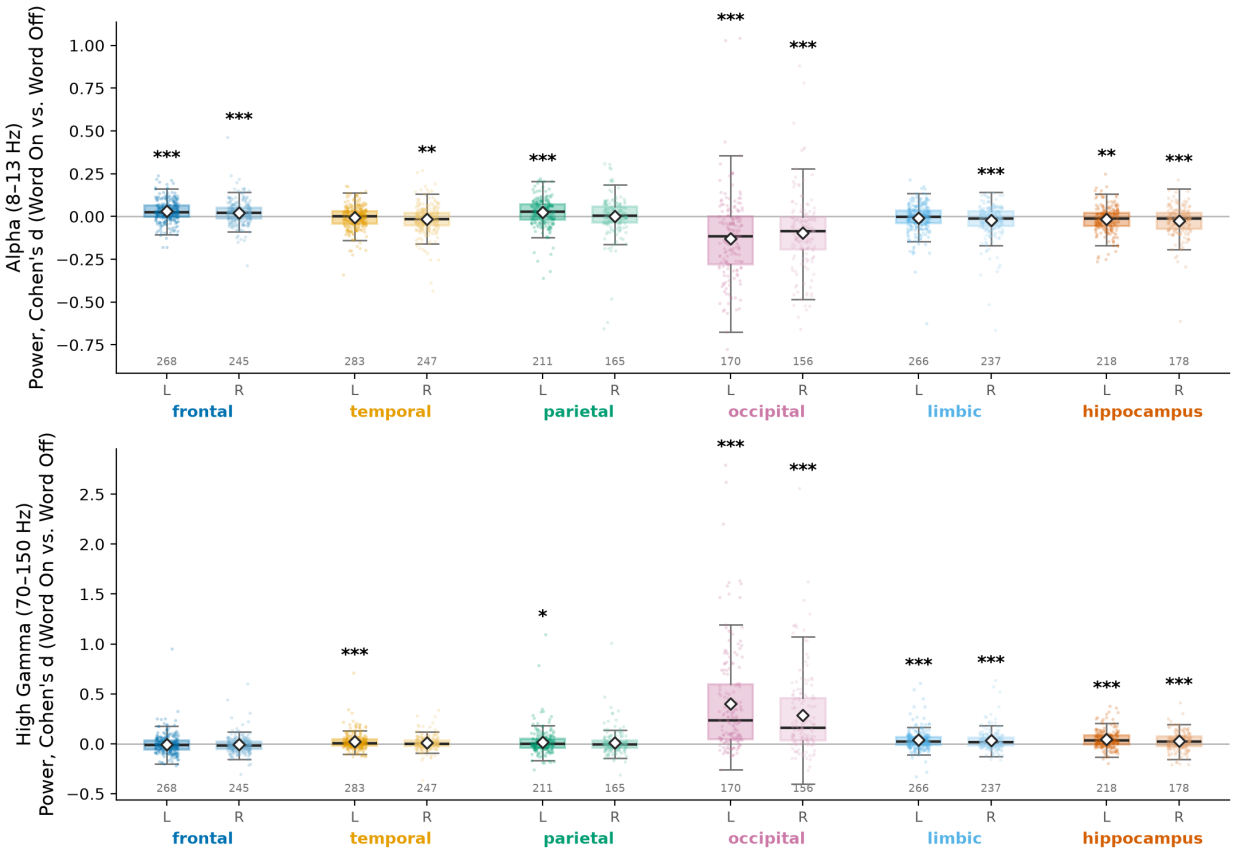


Figure 12: Power, subject distributions

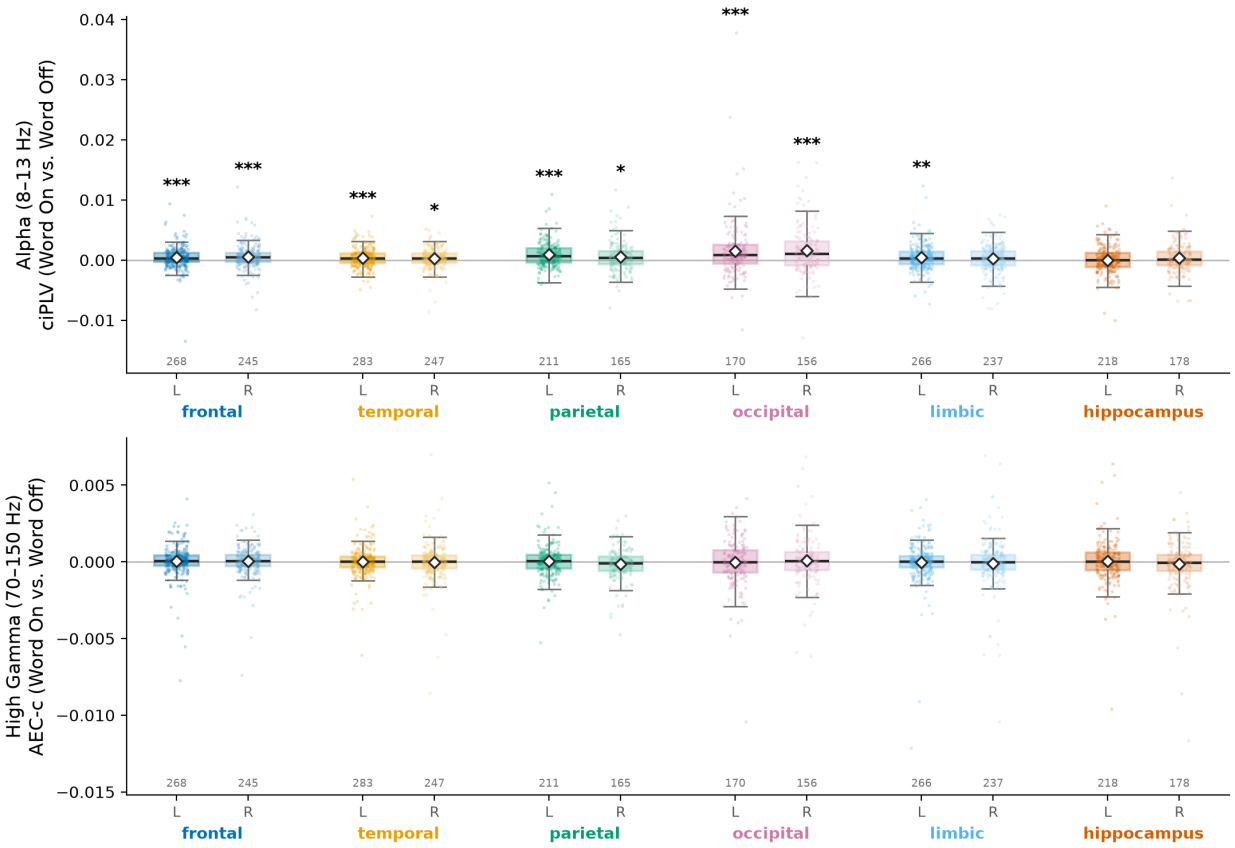


Figure 13: Synchrony, subject distributions

- Task-evoked activity inflates connectivity: Cole et al. 2019; van Diepen & Mazaheri 2018; Kramer et al. 2008 (sharp transients).
- Alternatives to subtraction:
  - Matched condition contrast with a label-shuffle null: Solomon et al. 2017; Maris et al. 2007.
  - Trial-shuffled (shift-predictor) surrogates: Perkel et al. 1967; Aertsen et al. 1989; Brody 1999.
  - Connectivity aligned to internal HFA peaks: Dede et al. 2025.
  - Phase consistency on the same data: Colyer & Kahana 2025.